

Altermagnetic piezomagnetism and spin-lattice coupling

Interplay of spin-orbit coupling, occupation, and structural degrees of freedom

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Research Focus & Methodology

We combine group theory and density functional theory (DFT) calculations to predict the piezomagnetic response of several collinear altermagnets. For the direct piezomagnetic effect, symmetry-breaking strains are applied and the piezomagnetic coefficients are extracted through finite differences as a function of spin-orbit coupling (SOC). The SOC is included through the standard second variational method, with λ_{SOC} denoting the prefactor of the SOC term in the Kohn-Sham Hamiltonian. For the inverse piezomagnetic effect, we apply a Zeeman magnetic field in the absence of SOC, focusing exclusively on the spin response and neglecting orbital contributions. Phonon calculations are carried out using the finite difference method.

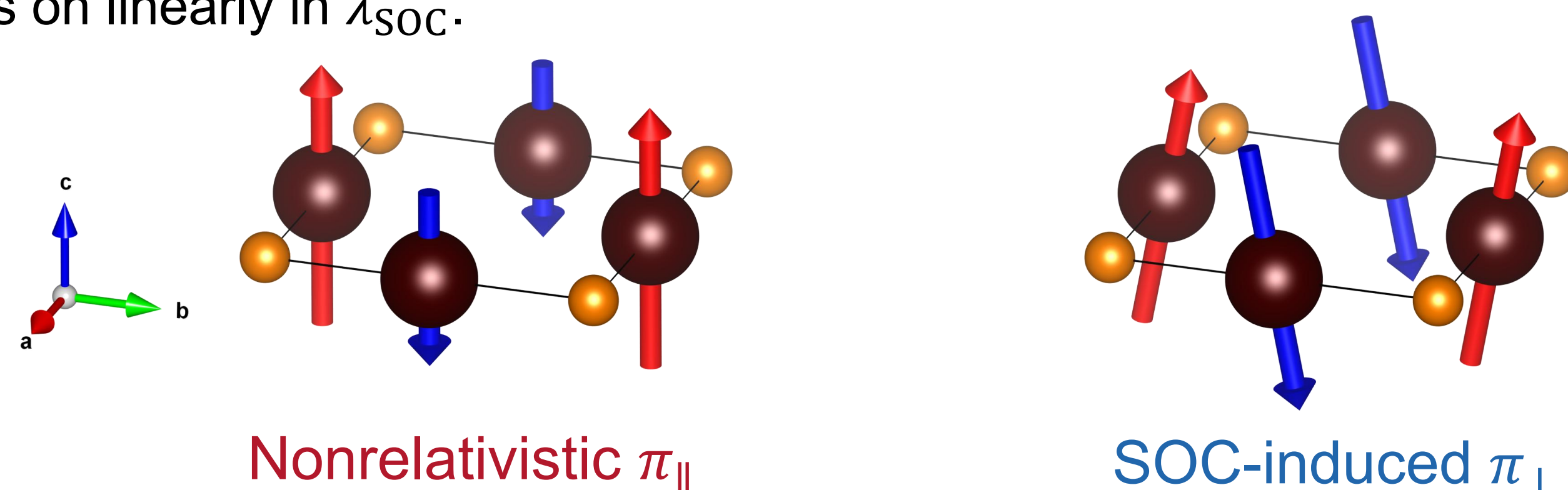


Direct Piezomagnetism and the role of SOC

We study the piezomagnetic response of several altermagnets, defined as the linear change in net magnetization of the system as a function of applied external strain:

$$M_i = \Pi_{ijk} \varepsilon_{jk}.$$

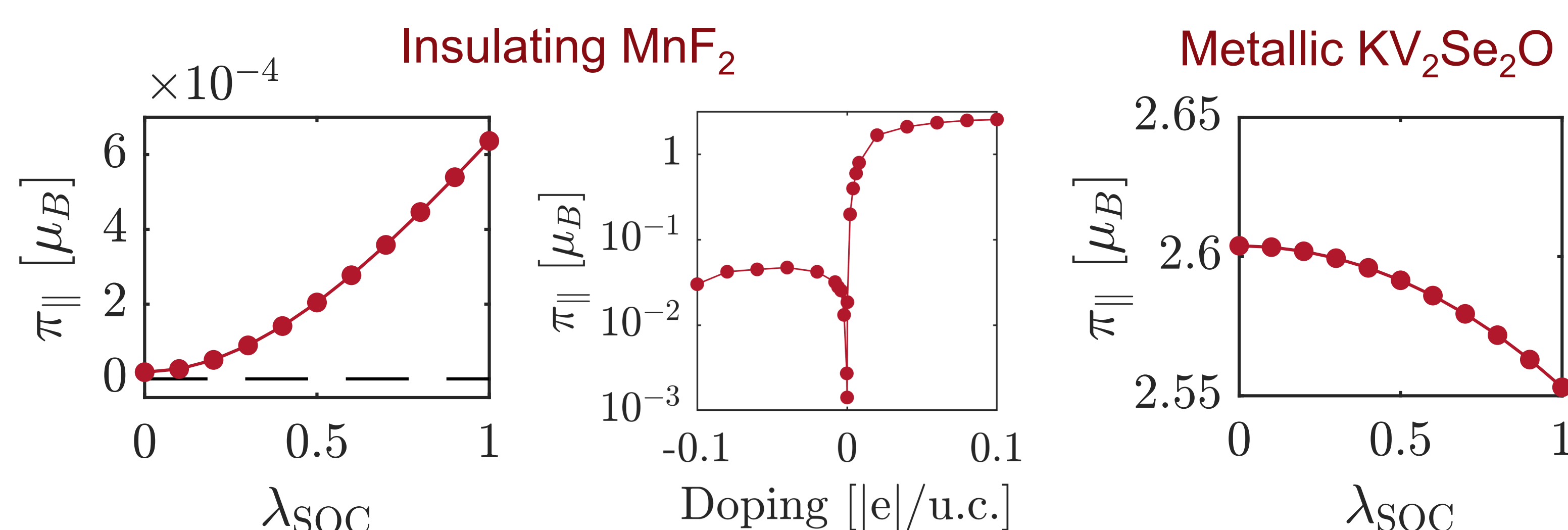
Piezomagnetic tensor components in **collinear altermagnets** fall into two **classes**: **nonrelativistic** $\pi_{\parallel}$, which survives at $\lambda_{\text{SOC}} = 0$, and **SOC-induced** $\pi_{\perp}$, which switches on linearly in λ_{SOC} .



Nonrelativistic piezomagnetism

In d-wave altermagnets, $\pi_{\parallel}$ is always symmetry-allowed and arises from strain-induced changes in band occupation. It is therefore expected to be strongly suppressed in **band insulators** at zero temperature. This is confirmed by our DFT calculations in MnF_2 , which has a large band gap of ~ 2 eV: in the absence of SOC the piezomagnetic response is negligible, and grows only at order λ_{SOC}^2 .

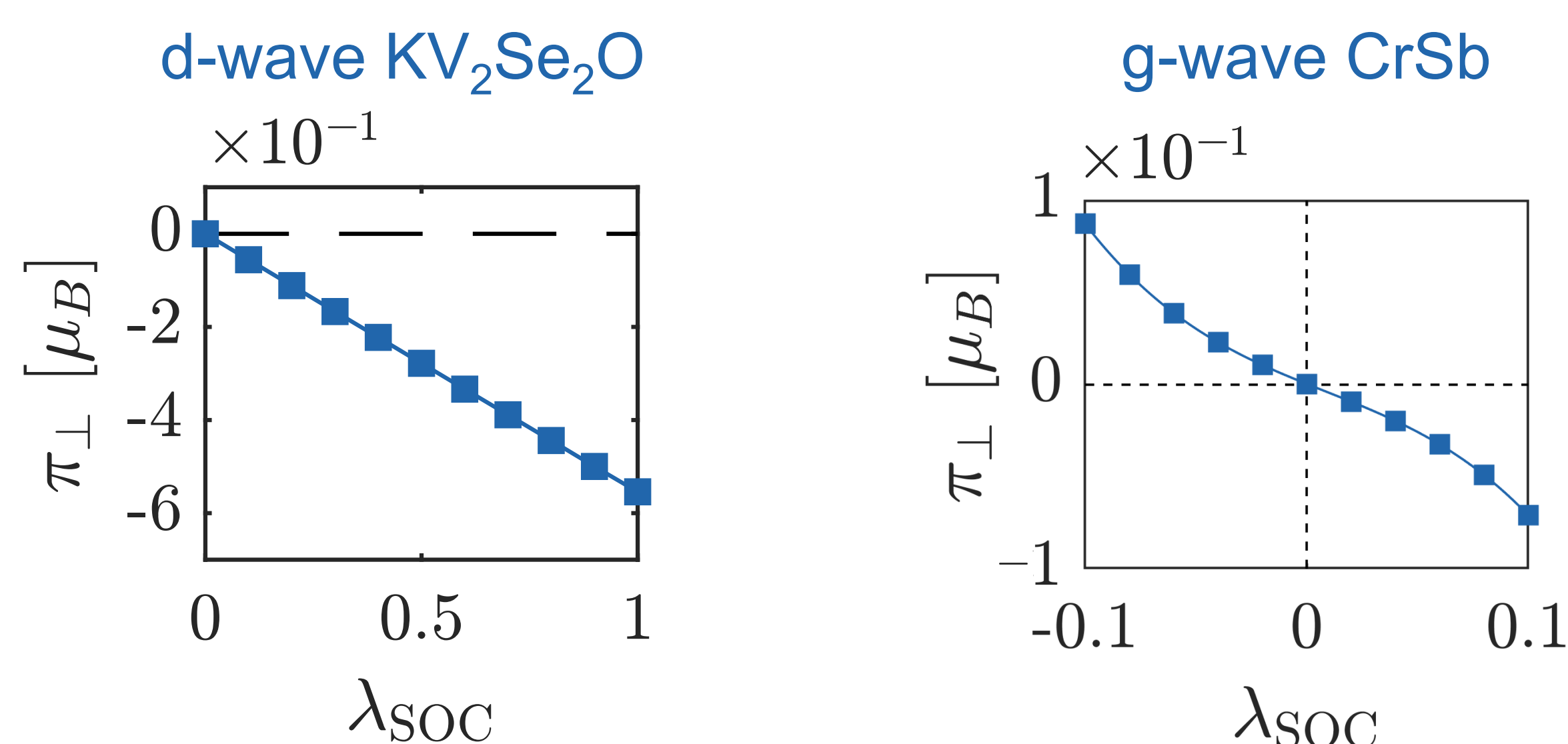
Importantly, even a small amount of charge carriers introduced through **doping** can **enhance** $\pi_{\parallel}$ by several orders of magnitude. In **metals** with significant Fermi-level DOS, $\pi_{\parallel}$ is expected to be large. This is illustrated by $\text{KV}_2\text{Se}_2\text{O}^{\dagger}$, where the piezomagnetic response is substantial already at $\lambda_{\text{SOC}} = 0$ and is renormalized only at quadratic order in SOC.



Piezomagnetic coefficients estimated from DFT

SOC-induced piezomagnetism

The occupation-driven $\pi_{\parallel}$ is symmetry-forbidden in g-wave altermagnets. However, a distinct SOC-enabled contribution, $\pi_{\perp}$, arising from strain-induced **tilting of the spins**, is symmetry-allowed in both d-wave and g-wave altermagnets and scales linearly with λ_{SOC} . As shown below, this linear scaling is clearly visible in both d-wave $\text{KV}_2\text{Se}_2\text{O}$ and g-wave CrSb . The linear regime is narrower in CrSb due to the larger SOC strength of the Sb ions.



Piezomagnetic coefficients estimated from DFT

L. Buiarelli, R.M. Fernandes, T. Birol, Phys. Rev. B **112**, 224442 (2025)

R. Ohlendorf et al., arXiv: 2601.19343 (2026)

[†] Chosen as a representative of the family, see: B. Thapa et al., arXiv: 2602.18672 (2026)

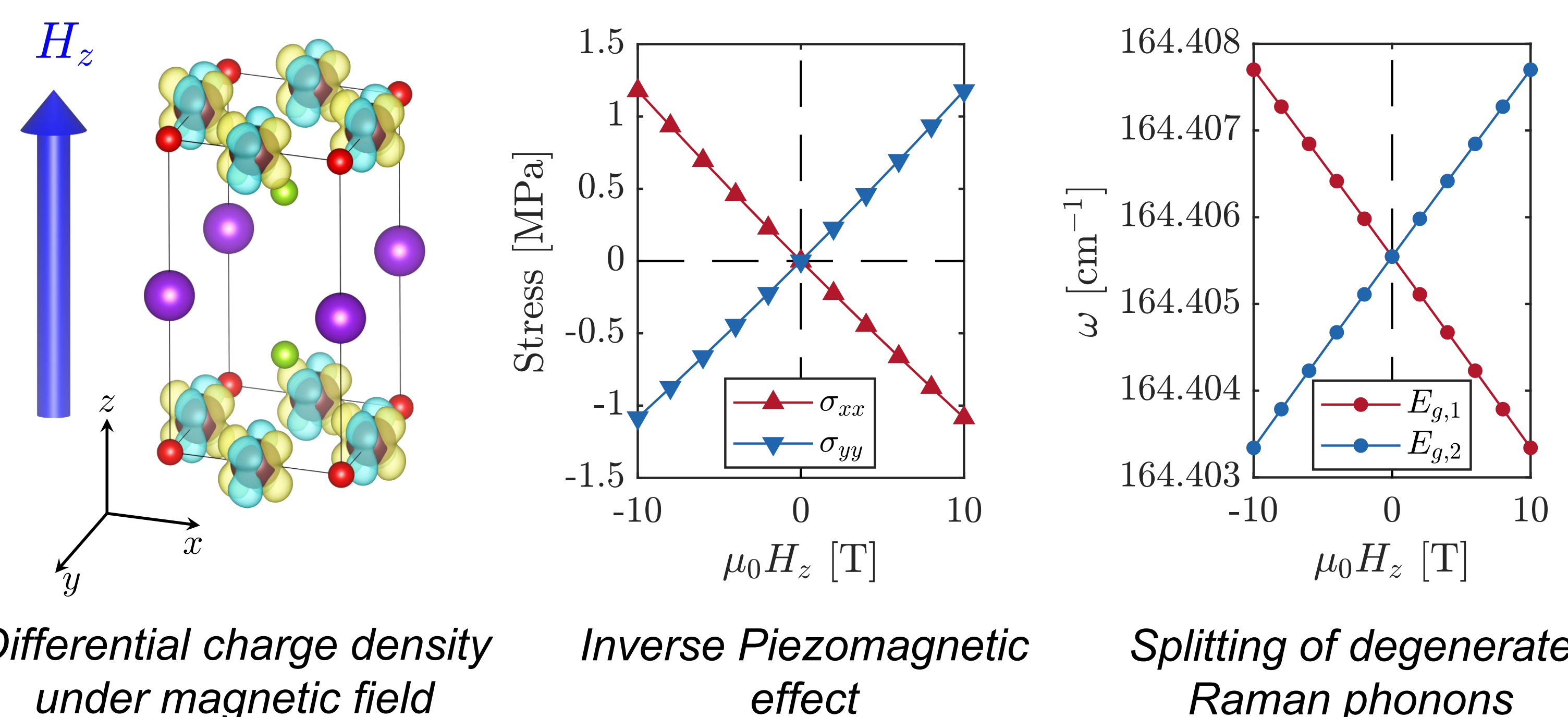
Inverse piezomagnetism w/o SOC

d-wave | splitting of Raman phonons

We now look at the inverse effect, where strain varies linearly with an applied external magnetic field:

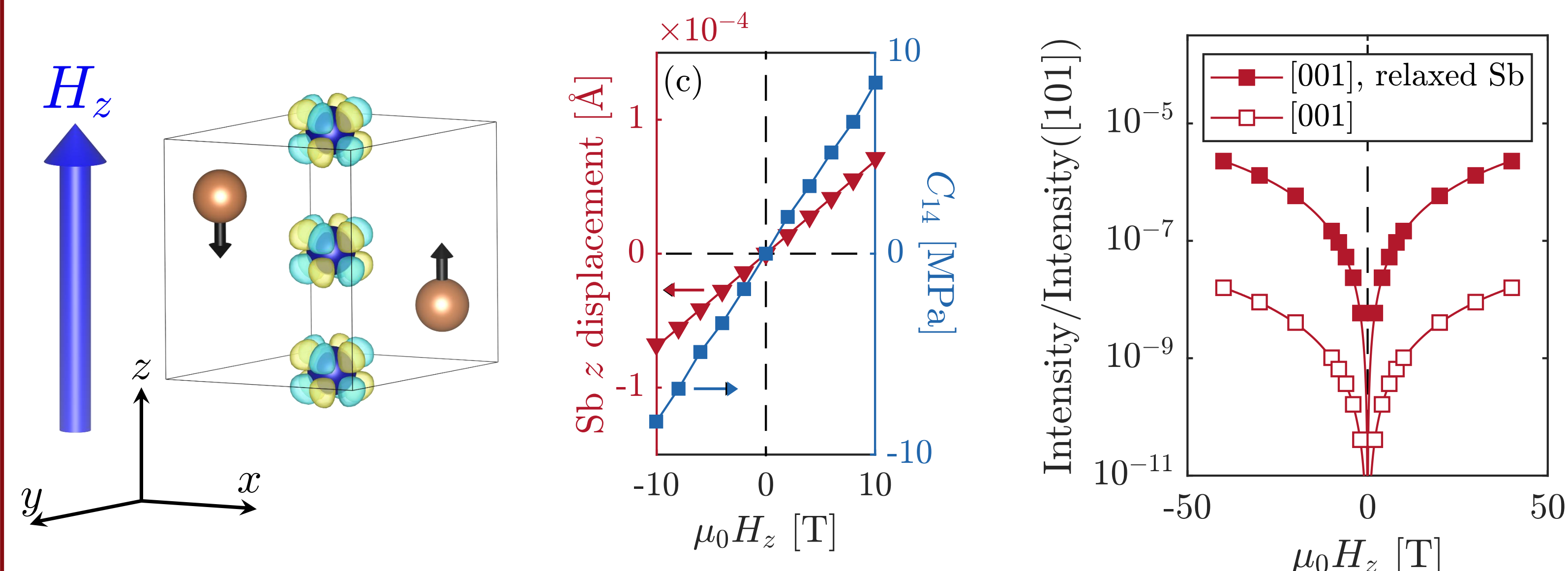
$$\varepsilon_{ij} = \Pi_{ijk} M_k.$$

Without SOC, this effect is unique to d-wave altermagnets, where it drives linear coupling between an external magnetic field and **charge quadrupolar responses**. We simulated $\text{KV}_2\text{Se}_2\text{O}^{\dagger}$, where signatures appear in the atomic **charge density** (a quadrupolar component proportional to $x^2 - y^2$), the **stress tensor** (a $\sigma_{xx} - \sigma_{yy}$ difference that drives orthorhombic distortion), and the **phonon spectrum** (splitting of degenerate Raman modes).



g-wave CrSb | new Bragg peaks under magnetic field

Without SOC, piezomagnetism is forbidden in g-wave altermagnets; instead, an external magnetic field couples linearly to **charge hexadecapolar responses** that break rotational symmetry. In CrSb , this manifests as a hexadecapolar component proportional to $yz(x^2 - 3y^2)$ in the Cr charge density and displacements of the Sb ions from their high-symmetry sites. The resulting loss of screw rotational symmetry activates forbidden (00L) peaks with odd L in **X-ray diffraction**, and new contributions to the **elastic constants** appear, accessible via ultrasound spectroscopy or acoustic birefringence.



Differential charge density under magnetic field Induced Sb displacement and new elastic constant Systematic absences vs external magnetic field

L. Buiarelli, R.M. Fernandes, T. Birol (in preparation)

Conclusions

Direct and inverse **piezomagnetism** possess the right symmetries to **distinguish altermagnets** from conventional antiferromagnets. They are always symmetry-allowed in d-wave altermagnets, whereas in g-wave altermagnets they are permitted only in the presence of spin-orbit coupling. Consequently, **probing the nonrelativistic order parameter of g-wave altermagnets** requires looking beyond strain and stress to more **generalized magneto-structural responses**. Here we propose that altermagnets exhibit a unique feature, rotational symmetry breaking under an external magnetic field, which can be probed via Raman spectroscopy, X-ray diffraction, and ultrasound spectroscopy.



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